

III.

$$\hat{y}_h = \hat{\beta}_0 + \hat{\beta}_1 d_h = \bar{y} + \hat{\beta}_1 (d_h - \bar{x})$$

$$= \sum_{j=1}^n \left(\frac{1}{n} + \frac{(d_h - \bar{x})(d_j - \bar{x})}{Sxx} \right) y_j$$

$$\mathbb{E}[y_h] = \beta_0 + \beta_1 d_h$$

$$\begin{aligned} \text{var}(\hat{y}_h) &= \sum_{j=1}^n \left(\frac{1}{n} + \frac{(d_h - \bar{x})(d_j - \bar{x})}{Sxx} \right)^2 \sigma^2 \\ &= \sigma^2 \left(\frac{1}{n} + \frac{(d_h - \bar{x})^2}{Sxx} \right) \end{aligned}$$

$$\frac{\hat{y}_h - \mathbb{E}(y_h)}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{(d_h - \bar{x})^2}{Sxx} \right)}} \sim N(0,1)$$

$$\frac{\hat{\sigma}^2}{\sigma^2} = \frac{SSE}{\sigma^2} / n-1 \sim \chi^2(n-1) / n-1$$

$$\frac{\hat{y}_h - \mathbb{E}(y_h)}{\sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(d_h - \bar{x})^2}{Sxx} \right)}} = \frac{\frac{\hat{y}_h - \mathbb{E}(y_h)}{\sqrt{\sigma^2 \left(\frac{1}{n} + \frac{(d_h - \bar{x})^2}{Sxx} \right)}}}{\frac{\sqrt{\hat{\sigma}^2}}{\sqrt{\sigma^2}}} \sim \frac{N(0,1)}{\chi^2(n-1)/n-1}$$

$$\left(\hat{\beta} \perp \hat{\sigma}^2 \text{ } \because \text{ } \sigma^2 (X^T X)^+ X^T (I - H_X) = 0 \right) \sim \pm(n-1)$$

2.

$$R^2 = \frac{SSR}{ST} = \frac{SSR}{SSR+SSE} \quad \& \quad (H_x - H_1)(I - H_x) \\ = H_x - H_x - H_1 + H_1$$

$$= \frac{\frac{SSR}{\cancel{Q^2}}}{\frac{SSR}{\cancel{Q^2}} + \frac{SSE}{\cancel{Q^2}}} = \frac{Q^2(p)}{Q^2(p) + Q^2(n-p-1)} = 0$$

$$= \frac{\Gamma(\frac{p}{2}, 2)}{\Gamma(\frac{p}{2}, 2) + \Gamma(\frac{n-p-1}{2}, 2)} \sim \text{beta}(\frac{p}{2}, \frac{n-p-1}{2})$$

□

[3].

$$A\beta = 0, AC = 0 \Rightarrow \beta = C\theta \quad \because \beta \in N(A)$$

$$\min_{\beta} (y - X\beta)^T (y - X\beta) + \lambda (A\beta - 0)$$

$$\Leftrightarrow \min_{\theta} (y - XC\theta)^T (y - XC\theta) = 2$$

$$\frac{\partial L}{\partial \theta} = -2C^T X^T (y - XC\theta) = 0$$

$$\Rightarrow C^T X^T y - (C^T X^T X C)\theta = 0$$

$$\Rightarrow \hat{\theta} = (C^T X^T X C)^{-1} C^T X^T y$$

full column rank
full row rank

$\Rightarrow C^T X^T X C$ is nonsingular

$$\Rightarrow \hat{\beta} = C(C^T X^T X C)^{-1} C^T X^T y$$

□

4.

$$y^* = B^T y = B^T X \beta + B^T \varepsilon \gamma + B^T \varepsilon = B^T \varepsilon \gamma + B^T \varepsilon$$

$$\mathbb{E} y^* = 0, \text{Var}(y^*) = B^T V B$$

$$y^* \sim N(0, B^T V B)$$

$$f(y^*) = \frac{1}{(2\pi)^{p/2} |B^T V B|} e^{-\frac{y^{*T} (B^T V B)^{-1} y^*}{2}}$$

$$\begin{aligned} \ell_R &= -\frac{p}{2} \log 2\pi - \frac{1}{2} \log |B^T V B| - \frac{1}{2} y^T B (B^T V B)^{-1} B^T y \\ &\propto -\frac{1}{2} y^T B (B^T V B)^{-1} B^T y - \frac{1}{2} \log |B^T V B| \end{aligned}$$

$$\begin{aligned} \frac{\partial \ell_R}{\partial x} &= \frac{1}{2} y^T B (B^T V B)^{-1} B^T \varepsilon \varepsilon^T B (B^T V B)^{-1} B^T y \\ &\quad - \frac{1}{2} \text{tr}((B^T V B)^{-1} B^T \varepsilon \varepsilon^T B) = 0 \end{aligned}$$

$$\Rightarrow \text{tr}(P \varepsilon \varepsilon^T) = y^T P \varepsilon \varepsilon^T P y$$

$$\text{where } P = \underbrace{B(B^T V B)^{-1} B^T}_L = \underbrace{V^{-1} - V^{-1} X (X^T V^{-1} X)^{-1} X^T V^{-1}}_R$$

$$B^T X = 0$$

$$\Rightarrow B(B^T V B)^{-1} B^T X = 0$$

LX

$$\& B(B^T V B)^{-1} B^T V B = B$$

$LV B$

$$V^{-1}X - V^{-1}X(X^T V^{-1}X)^{-1}X^T V^{-1}X = 0 \quad \text{RX}$$

$$V^{-1}VB - V^{-1}X(X^T V^{-1}X)^{-1}X^T B = B \quad \text{RVB}$$

$$LX = RX, \quad LVB = RVB$$

$$LIX \quad VBI = RIX \quad VBI$$

$$\text{Let } LIX \quad VBI \begin{pmatrix} a \\ c \end{pmatrix} = 0$$

$$\Rightarrow Xa + VBC = 0$$

$$B^T Xa + B^T VBC = 0 \Rightarrow \underbrace{B^T VBC}_{\text{Singular}} = 0$$

$$\Rightarrow C = 0$$

$$Xa = 0 \text{ \& } X: \text{full column rank.}$$

$$\Rightarrow a = 0$$

$$\text{So } LIX \quad VBI \text{ invertible}$$

$$\Rightarrow L = R$$

$$\text{i.e. } B(B^T V B)^{-1} B^T = V^{-1} - V^{-1}X(X^T V^{-1}X)^{-1}X^T V^{-1}$$